

Eenheid 4.4

Note Title

2015/03/26

Onbepaalde vorme en L'Hospital se Reël. (Stewart p 304)

• Vooraf:

① $\lim_{x \rightarrow 3} \frac{x-3}{x^2-9}$ $\left(\frac{0}{0} \right)$ ^{Vorm}

$$= \lim_{x \rightarrow 3} \frac{x-3}{(x-3)(x+3)} = \frac{1}{6}$$

Plan: Faktoriseer

• ② $\lim_{x \rightarrow \infty} \frac{x^5 + 5x^3}{x^4 + x^2}$ $\left(\frac{\infty}{\infty} \right)$

Plan: Deel bo en onder
met hoogste mag van
x onder die lyn

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^5}{x^4} + \frac{5x^3}{x^4}}{1 + \frac{1}{x^2}} \begin{matrix} \sim \infty \\ \sim 0 \end{matrix}$$
$$= \lim_{x \rightarrow \infty} \frac{x + \frac{5}{x}}{1} = \infty$$

• Onbepaalde vormen is limiet van die
vormen:

① $\frac{0}{0}$; $\frac{\pm\infty}{\pm\infty}$

② $\infty - \infty$

③ $0 \times \pm\infty$

④ $0^0, \infty^0, 1^\infty$
... ..

L' Hospital se Reël

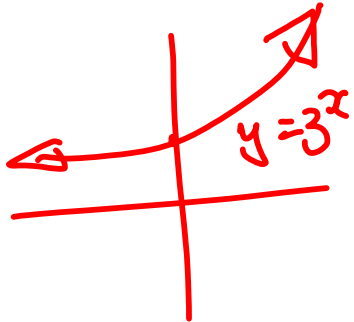
Als $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ 'n limiet is van die
onbepaalde vorm $\frac{0}{0}$ of $\frac{\pm\infty}{\pm\infty}$,
dan is

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \overset{\text{VORM}}{=} \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

VOORBEELDE:

Vorm 1 : $\left(\frac{0}{0}\right)$ en $\left(\frac{\pm\infty}{\pm\infty}\right)$

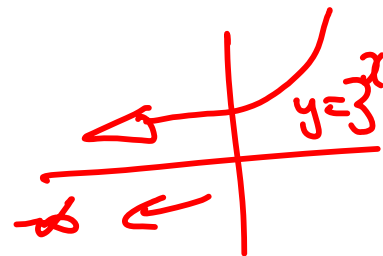
$$\textcircled{1} \lim_{x \rightarrow 3} \frac{x-3}{x^2-9} \left(\frac{0}{0}\right) \stackrel{H}{=} \lim_{x \rightarrow 3} \frac{1}{2x} = \frac{1}{6}$$

$$\textcircled{2} \lim_{x \rightarrow \infty} \frac{3^x}{x^2} \left(\frac{\infty}{\infty}\right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{3^x \ln 3}{2x} \left(\frac{\infty}{\infty}\right)$$
$$\stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{(\ln 3)^2 (3)^{x+1}}{2} = \infty$$


$$\begin{aligned} (\ln 3)(3^x) &= y \\ y' &= (\ln 3) \times 3^x \ln 3 \end{aligned}$$

$$\textcircled{3} \lim_{x \rightarrow -\infty} \frac{3^x}{x^2} \left(\frac{0}{\infty}\right) = 0$$

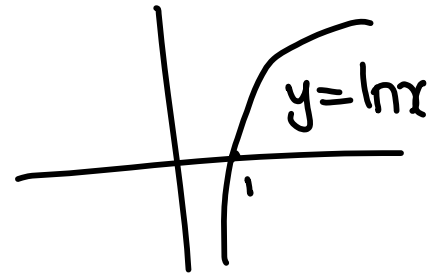
kan nie L'Hospital gebruik nie



$$\textcircled{4} \lim_{x \rightarrow \infty} \frac{\ln x}{x^2} \left(\frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{2x} = \lim_{x \rightarrow \infty} \frac{1}{2x^2} = 0$$

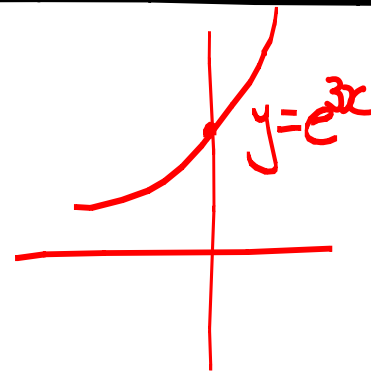
$$\textcircled{5} \lim_{x \rightarrow -\infty} \frac{\ln x}{x^2}$$

Limiet bestaat nie
 $f(x) = \ln x$
 $D_f = (0, \infty)$



$$\textcircled{6} \lim_{x \rightarrow 0} \frac{2x}{e^{3x} - 1} \left(\frac{0}{0} \right)$$

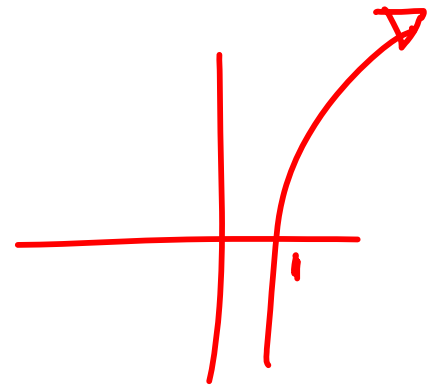
$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{2}{3e^{3x}} = \frac{2}{3}$$



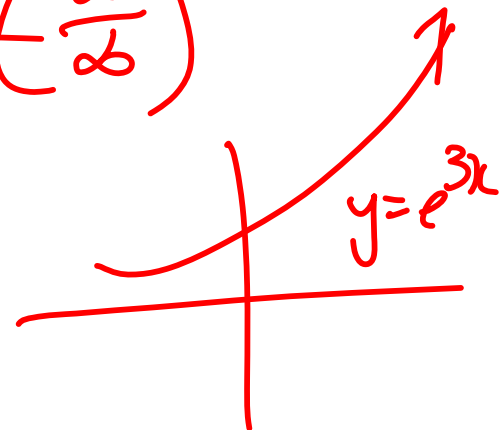
- VORM
- +1
- Vereenvoudig
- +1 ?? • antw

$$\textcircled{7} \lim_{x \rightarrow \pi/2} \frac{\cos x}{2x - \pi} \left(\frac{0}{0} \right) \stackrel{H}{=} \lim_{x \rightarrow \pi/2} \frac{-\sin x}{2} \\ = -\frac{1}{2}$$

$$\textcircled{8} \lim_{x \rightarrow \infty} \frac{x \ln x}{x + \ln x} \left(\frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{\ln x + x \cdot \frac{1}{x}}{1 + \frac{1}{x}} \\ = \lim_{x \rightarrow \infty} \frac{\ln x + 1}{1 + \frac{1}{x}} \rightarrow 0 \\ = \infty \text{ (antwoord)} \quad \therefore \text{Limiet DNE}$$



$$\textcircled{9} \lim_{x \rightarrow \infty} \frac{x^2}{e^{3x} + 4} \left(\frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{2x}{3e^{3x}} \left(\frac{\infty}{\infty} \right) \\ \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{2}{(3)(3)e^{3x}} \\ = 0 \checkmark$$



VORM 2: $\infty - \infty$

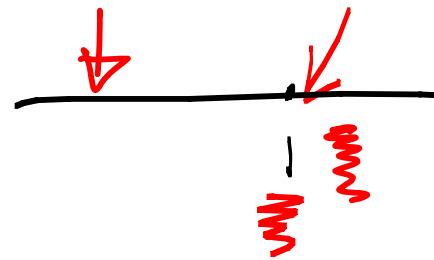
Plan: Robeer
Faktoriseer
om teken te kry

$$\begin{aligned} \textcircled{1} \lim_{x \rightarrow \infty} x^3 - x^4 & \quad (\infty - \infty) \\ & \quad \begin{matrix} (+) & \times & (-) \\ x^3 & (1-x) \end{matrix} \\ & = \lim_{x \rightarrow \infty} x^3 (1-x) = \frac{?}{?} \\ & \quad \text{(teken?)} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \lim_{x \rightarrow -\infty} x^4 + x^3 & \quad ((-\infty)^4 + (-\infty)^3) = (\infty - \infty) \rightarrow \text{Faktoriseer!} \\ & = \lim_{x \rightarrow -\infty} x^3 (x+1) = +\infty \\ & \quad \begin{matrix} (-) & \times & (-) \end{matrix} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \lim_{x \rightarrow \infty} x^3 + x^4 & \quad (\infty + \infty) \quad \text{Nie in onbepaalde vorm} \\ & = \infty \end{aligned}$$

$$\textcircled{4} \lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right) \quad (\infty - \infty)$$



Plan: Gemeenschappelijke noemer

$$= \lim_{x \rightarrow 1^+} \frac{(\cancel{x-1}) - (\ln \cancel{x})}{(\ln x) \cancel{x} (x-1)} \quad \left(\text{Vorm: } \frac{0}{0} \right) \checkmark$$

$$\textcircled{H} = \lim_{x \rightarrow 1^+} \frac{1 - \frac{1}{x}}{\left(\frac{1}{x}\right)(x-1) + (\ln x)(1)}$$

$$= \lim_{x \rightarrow 1^+} \frac{\frac{x-1}{x}}{(x-1) + x \ln x}$$

$$= \lim_{x \rightarrow 1^+} \frac{x-1}{x-1 + x \ln x} \quad \left(\frac{0}{0} \right) = \lim_{x \rightarrow 1^+} \frac{1}{1 + (\ln x + x \cdot \frac{1}{x})} = \frac{1}{1+1} = \frac{1}{2}$$

- VORM ??
- H
- Vereenvoudig
- Vorm
- H
- antw.

VORM 3: ($\pm \infty \times 0$)

$$\textcircled{1} \lim_{x \rightarrow 0^+} x \ln x$$

$\overset{+0}{x} \quad \overset{-\infty}{\ln x}$
Vorm ($0 \times \infty$)

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{x} \times -x^2 = \lim_{x \rightarrow 0^+} -x = 0$$

Plan: vorm: produkt

Dann? Bruch
 $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \left(\frac{0}{0} \right) \left(\frac{\pm \infty}{\pm \infty} \right) \stackrel{H}{=}$

$$\textcircled{2} \lim_{x \rightarrow \infty} x^3 e^{-x^2} (\infty \times 0)$$

$$\stackrel{!}{=} \lim_{x \rightarrow \infty} \frac{x^3}{e^{x^2}} \left(\frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{3x^2}{e^{x^2} \times 2x} \quad \text{(Vereenvoudig)}$$

$$= \lim_{x \rightarrow \infty} \frac{3x}{2e^{x^2}} \left(\frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{3}{2xe^{x^2} \times 2x}$$

$$= \lim_{x \rightarrow \infty} \frac{3}{4xe^{x^2}} = 0$$

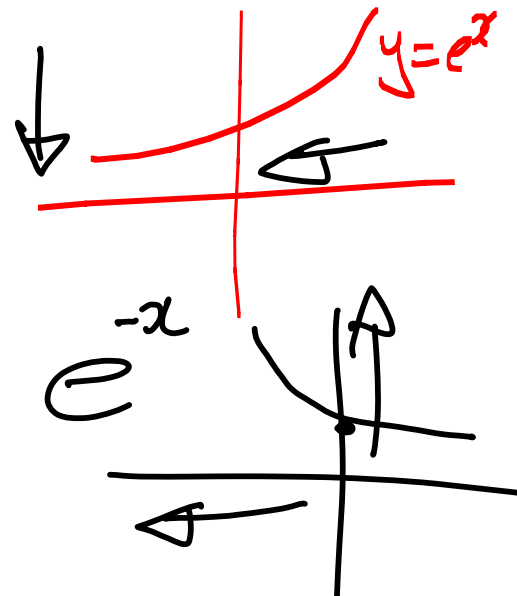
$$\textcircled{3} \lim_{x \rightarrow -\infty} x e^x (-\infty \times 0)$$

$$= \lim_{x \rightarrow -\infty} \frac{x}{\frac{1}{e^x}} \left(\frac{-\infty}{+\infty} \right)$$

$$\stackrel{H}{=} \lim_{x \rightarrow -\infty} \frac{1}{-e^{-x}} = 0$$

produkt
↓
Breuk

$$\frac{1}{e^x} = e^{-x}$$



VORM 4: Vir die vorms $0^0, \infty^0$ en 1^∞
NB gebruik die ln funksie

Plan: vir $0^0, \infty^0, 1^\infty$ ✓

- Stel $y = f(x)^{g(x)}$
- Gebruik ln: $\ln y = \ln f(x)^{g(x)}$
- Eienskappe van ln
- $\ln y = g(x) \ln f(x)$ ($0 \times \infty$)
- Skryf f as i breuk
- VORM $\left(\frac{0}{0}\right) \left(\frac{\pm \infty}{\pm \infty}\right)$
- Gebruik L'Hospital

~~NB~~ antwoord $\lim_{x \rightarrow \infty} \cancel{y}$

① $\lim_{x \rightarrow \infty} \left(1 + \frac{4}{x}\right)^{3x}$
 Vorm: $(1)^{\infty}$ ✓

Stel $y = \left(1 + \frac{4}{x}\right)^{3x}$

$\Rightarrow \ln y = \ln \left(1 + \frac{4}{x}\right)^{3x} = 3x \times \ln \left(1 + \frac{4}{x}\right)$
 $\Rightarrow \lim_{x \rightarrow \infty} \ln y \Rightarrow \lim_{x \rightarrow \infty} 3x \times \ln \left(1 + \frac{4}{x}\right)$ ($\infty \times 0$)
 (product)

$= 3 \lim_{x \rightarrow \infty} x \times \ln \left(1 + \frac{4}{x}\right)$

$= 3 \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{4}{x}\right)}{\frac{1}{x}}$ ($\frac{0}{0}$)

Break
 $\downarrow$
 $\neq$

$\stackrel{H}{=} 3 \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{4}{x} \times -\frac{4}{x^2}}$

Verenwoordig

$= 12 \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{4}{x}} = 12$

Antwoord: $\lim_{x \rightarrow \infty} \ln y = 12$
 $\lim_{x \rightarrow \infty} y = e^{12}$

② $\lim_{x \rightarrow \infty} (e^x + 6x)^{\frac{2}{x}}$ Norm: (∞^0)

Let $y = (e^x + 6x)^{\frac{2}{x}}$

$\Rightarrow \ln y = \ln(e^x + 6x)^{\frac{2}{x}} = \frac{2}{x} \times \ln(e^x + 6x)$

$\Rightarrow \lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{2}{x} \times \ln(e^x + 6x)$ (Norm $(0 \times \infty)$ $\downarrow$ Breuk)

$= 2 \lim_{x \rightarrow \infty} \frac{\ln(e^x + 6x)}{x}$ Norm $(\frac{\infty}{\infty})$

$\stackrel{H}{=} 2 \lim_{x \rightarrow \infty} \frac{\frac{1}{e^x + 6x} \times (e^x + 6)}{1}$

$= 2 \lim_{x \rightarrow \infty} \frac{e^x + 6}{e^x + 6x} \left(\frac{\infty}{\infty}\right)$

✓
Breuk

$$\stackrel{H}{=} 2 \lim_{x \rightarrow \infty} \frac{e^x}{e^x + 6} \left(\frac{\infty}{\infty} \right)$$

$$\stackrel{H}{=} 2 \lim_{x \rightarrow \infty} \frac{\cancel{e^x}}{\cancel{e^x}} \quad \text{Vereenvoudig} = 2 \times 1 = 2$$

$$\therefore \lim_{x \rightarrow \infty} \ln y = 2 \Rightarrow \lim_{x \rightarrow \infty} y = e^2$$

(3) $\lim_{x \rightarrow 0^+} x^{\frac{1}{\ln x}}$ Vorm (0^0) (Gebruik $\ln$)

$$\text{Set } y = x^{\frac{1}{\ln x}}$$

$$\Rightarrow \ln y = \ln x^{\frac{1}{\ln x}} = \frac{1}{\cancel{\ln x}} \times \cancel{\ln x} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \ln y = 1 \quad \text{Maar} \quad \lim_{x \rightarrow 0^+} y = e^1$$

* Praktikum # 55

$$\lim_{x \rightarrow \infty} (x - \ln x) \text{ Faktoriser?}$$

($\infty - \infty$)

$$= \lim_{x \rightarrow \infty} x \left(1 - \frac{\ln x}{x} \right)$$

$$= \lim_{x \rightarrow \infty} x \left(\lim_{x \rightarrow \infty} 1 - \lim_{x \rightarrow \infty} \frac{\ln x}{x} \right)$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} \left(\frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0$$

$$\text{Dann } \lim_{x \rightarrow \infty} x (1 - 0) = \infty$$